

Aaryan Desai

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SUMMARY

Mechanical Engineer specializing in robotics, automation, and mechatronic systems. Experienced in CAD/CAE, ROS/ROS2, PLC programming, and sensor integration. Skilled in autonomous system design and development, with a focus on scalable, efficient robotic solutions.

EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Expected May 2028

Master of Science in Robotic Systems Development

Purdue University, West Lafayette, IN

Dec 2023

Bachelor of Science in Mechanical Engineering, Minor in Computer Science

GPA: 3.45/4.0 (Major GPA: 3.7/4.0)

Dean's List: Spring 2020 – Spring 2022; Semester Honors: Spring 2021 – Fall 2021, Fall 2023

Relevant Coursework: Autonomous Systems, Mechatronics, Control Systems, Machine Design, Data Structures, Senior Design

TECHNICAL SKILLS

Robotics/Automation: ROS/ROS2, PLC (Siemens/Allen-Bradley), Gazebo, Arduino, Ignition

CAD/CAE: CATIA V5/3DEXPERIENCE, SOLIDWORKS, Inventor

Programming: Python, C/C++, Java, MATLAB, SQL

Expertise: Sensor Integration, Control Systems, Mechatronic Design, Process Automation, Tooling and Fixture Design

EXPERIENCE

Manufacturing Engineer, Modineer, Niles, MI

Apr 2024 – Present

- Led robotic weld cell consolidation, which increased workflow efficiency and production capacity by **12%**.
- Optimized die transfer motion profile, reducing cycle time by **35%** (1.7s → 1.1s) and eliminated jitter.
- Built a Raspberry Pi and PLC data acquisition pipeline using Pycomm3; visualized cycle times and fault analytics in Power BI for real-time visibility.
- Designed die components, precision fixtures, and tooling to ensure repeatable robotic operations for complex geometries.

Production Control Equipment Engineering Intern, Tesla Inc., Austin, TX

May 2023 – Aug 2023

- Designed mechanical racks and dollies to streamline equipment setup and material flow on the production floor.
- Standardized team equipment selection and design, establishing modular, reusable production systems.

Production Control Equipment Engineering Intern, Tesla Inc., Fremont, CA

May 2022 – Nov 2022

- Delivered 24+ material handling solutions and automated data collection systems supporting a production ramp from 5,600 to 12,000 vehicles per week.
- Prototyped a low-fidelity, proof-of-concept universal robotic end-of-arm tool projected to save **\$300K** over two years.
- Integrated Siemens and Allen-Bradley PLCs over Ethernet/PROFINET networks; the setup was later used to train Controls Technicians on troubleshooting cross-network devices such as conveyor VFDs.

PROJECTS

Vision Inspection System, Modineer

May 2025 – Aug 2025

- Automated multi-feature part inspection, reducing cycle time by **43%** (from 83s to 47s) using integrated vision, proximity, and laser sensors.
- Designed and implemented the fixture, control panel, and relay-based automation, demonstrating a scalable mechatronic solution.

ME 588 Mars Rover, Purdue University

Jan 2023 – May 2023

- Led electronics, sensor integration, and programming for autonomous rover navigation and object manipulation.
- Developed autonomous algorithms for foam-cube detection, pickup, and placement, achieving 95% success rate in arena trials.
- Collaborated on mechanical design and control system integration, delivering fully autonomous operation.